

System and Network Administration and Data Communication and Networking

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Principles of Network and System Administration

-System is referred to the operating system of a computer and often, collectively the set of all computers that cooperate in a network.

3 main components:

Humans: using and running the fixed infrastructure

Host computers: devices that run software.

Network hardware: specialized devices.

-Shells can be used to write simple programs called scripts/batch files.

-In UNIX, the privileged account is called root.

- All file systems on Unix-like operating systems are based on a system of index nodes.

-The top or start of the Unix file tree is called the root filesystem '/'.

Network Troubleshooting Commands in Linux

- `ifconfig`: The interface configurator command is used to initialize an interface, and assign an IP address to the interface enabled or disabled.
- `ping`: command is the best way to test connectivity between two nodes.
- `traceroute`: command is used to show the number of hops taken to reach destination that determines travelling-path.
- `netstat`: command display connection info, routing table info.
- `dig`: command/query DNS related information like aRecord, CNAME, MX Record e.t.c.
- `nslookup`: command also use to find out DNS related query.
- `host`: helps to find name to IP or IP to name in IPV4 or IPV6.
- `tcpdump`: The Dumps Ethernet Packet

Activity to console, Showing Traffic.

DATA COMMUNICATIONS

- Network Reliability: is a measurement by the frequency of failure
- Network Security: is an issuing that includes protecting data from Unauthorized Access
→ Protecting data from damaged development.

Types of Communication

- A link is a communications pathway that transfers data from one device to another.

There are 2 types of connections:

- 1) Point-to-Point: provides a dedicated link between two devices.
- 2) Multipoint: provides which has more than two specific devices share a single link.

Physical Topology

- Physical Topology is the way in which a network is laid out physically
- The topology of a network is the geometric representation of the relationship of all the links linking devices (usually called nodes) to one another.

- 1) Mesh Topology: Every device has a dedicated point-to-point link to every other device, dedicated means that the link carries traffic only between the two devices it connects.
- 2) Star Topology: Each device has a dedicated point-to-point link only to a central controller, usually called a hub.
- 3) Bus Topology: The long cable acts as a backbone to link all the devices in a network.
- 4) Ring Topology: Each device has a dedicated point-to-point connection with

only the two devices on either side of it.

The effectiveness of data communications four fundamental characteristics:

- 1) Delivery: The system must deliver data to the correct destination.
- 2) Accuracy: The system must deliver data accurately.
- 3) Timeliness: The system must deliver data in a timely manner.
- 4) Jitter: refers to the variation in the packet arrival time.

Components

A data communications system has five components:

- 1) Message: The message is the information (data) to be communicated.

2) Sender: The sender is the device that sends the data message.

3) Receiver: The receiver is the device that receives the data message.

4) Transmission Medium: The transmission medium is the physical path by which the message travels from sender to receiver.

5) Protocol: A protocol is a set of rules that govern data communications.